Class Summary

Key question: Does motion need a force to maintain it?

Evidence: less resistance produces a longer travel distance.

Reasoning: if resistance is zero, speed does not need to decrease.

Newton’s First Law

An object remains at rest or in uniform straight-line motion unless a nonzero net force acts on it.
Balanced forces do not change the motion state.

Inertia

Inertia is the property of keeping the original motion state.

All objects have inertia.

Greater mass means greater inertia.

Experiment Conclusions

Rough surface: friction causes clear deceleration.

Smooth surface: friction is smaller, so motion lasts longer.

Air track: net force is nearly zero, so motion is nearly uniform.

Apply The Idea

When analyzing motion, ask:

1. What forces act on the object?
2. Are the forces balanced?
3. Is the net force zero or nonzero?